

Solutions and R Programming Examples: Confidence Interval

1.Fast Food

1.1) 7.093 1.2) 0.363 1.3) 0.779

1.4) (6.315,7.872)

We are 95% confident that the true population mean of the amount spent for lunch at a fast-food restaurant is between \$6.315 and \$7.872.

R Program

```
food<-c(7.42,6.29, 5.83, 6.50,8.34, 9.51,7.10, 6.80,5.90, 4.89,6.50, 5.52, 7.90, 8.30, 9.60)
```

```
m<- mean(food)
```

```
m
```

```
[1] 7.093333
```

```
sd<-sd(food)
```

```
n<-15
```

```
se<- sd/sqrt(n)
```

```
se
```

```
[1] 0.3630357
```

```
moferr<-qt(0.975,n-1)*se
```

```
moferr
```

```
[1] 0.7786341
```

```
lcl<-m-moferr
```

```
ucl<-m+moferr
```

```
lcl
```

```
[1] 6.314699
```

ucl

[1] 7.871967

.....

2. The complaints

2.1) 43.040 2.2) (31.125, 54.955)

.....

3. Flu shot

3.1) 0.00615 3.2) 0.00125 3.3) 0.0025 3.4) (0.0037, 0.00861) 3.5) Yes

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4. Basic life

4.1) 0.54255, 0.61381 4.2) (-0.12712, -0.01540)

R Program

```
prop.test(x=c(306,391), n=c(564,637),conf.level=0.95)
```

.....

2-sample test for equality of proportions with continuity correction

data: c(306, 391) out of c(564, 637)

X-squared = 5.9483, df = 1, p-value = 0.01473

alternative hypothesis: two.sided

95 percent confidence interval:

-0.12878962 -0.01373351

sample estimates:

prop 1 prop 2

0.5425532 0.6138148

5. Study time

5.1) 2.567 5.2) (-3.110, 8.244) We are 95% confident that the true population mean of study time per week between the two groups (students who planned to go to graduate school and those who did not) is between -3.110 – 8.244 hours.

This means that there is no statistically significant difference in the true mean of study time per week between the two groups.

R Program

```
plan<-c(15,7,15,10,5,5,2,3,12,16,15,37,8,14,10,18,3,25,15,5,5)
```

```
N_plan<-c(6,8,15,6,5,14,10,10,12,5)
```

```
t.test(plan,N_plan,conf.level=0.95,var.equal=TRUE)
```

.....

Two Sample t-test

data: plan and N_plan

t = 0.92469, df = 29, p-value = 0.3628

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-3.110283 8.243616

sample estimates:

mean of x mean of y

11.66667 9.10000

.....

6. Multiple myeloma

6.1) 86.143 6.2) (-28.261, 200.547) We are 95% confident that the true population mean difference in bone marrow microvessel density of patients who received transplants of stem cells (before and after) is between -28.261, 200.547.

This means that there is no statistically significant difference in the true mean of bone marrow microvessel density between the two groups.